

Chapter 9 - The Voodoo 3D Rasteriser

Voodoo is Intuition Engine's hardware triangle rasteriser. It draws flat-shaded, Gouraud-shaded, textured, depth-tested, alpha-tested, fogged, chroma-keyed, and dithered triangles into an RGBA framebuffer with an associated Z buffer. Voodoo is compositor layer 20, so its visible pixels sit above the other picture sources.

The programmer's path is the same as the rest of this guide: type BASIC commands first, then use POKE, PEEK, POKE8, and PEEK8 when direct register control is useful. The register names keep their Voodoo heritage, but the addresses, formats, side effects, and examples below are the Intuition Engine contract.

9.1 What Voodoo can show

Use Voodoo when a picture is easier to describe as triangles than as tiles or rectangles: a spinning sign, a shaded cockpit panel, a textured floor, a depth-tested overlay, or a sprite-like quad that needs alpha. It is still one card on the same bus. BASIC sets up the state, vertices become fixed-point register values, and the compositor places the finished frame above the other display chips.

Item	Value
Output	RGBA framebuffer with depth buffer
Default resolution	640 by 480
Maximum resolution	800 by 600
Primitive	Triangle, defined by three vertices
Per-vertex data	Position, RGBA, Z, S, T, and W
Texture memory	64 KB at \$D0000
Texture sizes	up to 256x256 in the texture memory
Texture formats	paletted, intensity, alpha, and ARGB
Depth functions	8 functions, numbered 0 to 7
Alpha test/blend	Yes
Fog	Per-pixel depth fog
Chroma key	Yes
Dither	Ordered 4x4 or 2x2

9.2 Programming model

Voodoo is state-machine driven:

1. Enable the chip with VOOD00 ON.
2. Set the picture size with VOOD00 DIM.
3. Set draw state, usually FBZ_MODE, ALPHA_MODE, and FBZCOLOR_PATH.
4. Clear the back buffer with VOOD00 CLEAR.

5. Write vertices with VERTEX A, VERTEX B, and VERTEX C.
6. Write colour, depth, texture, or W values with V00D00 TRI . . . commands.
7. Submit the triangle with TRIANGLE.
8. Publish the frame with V00D00 SWAP.

The simplest triangle is a typed BASIC program:

```
10 REM V00D00 FIRST TRIANGLE
20 V00D00 ON
30 V00D00 DIM 640,480
40 POKE32 &H000F8110,&H00008200
50 V00D00 CLEAR &HFF101020
60 VERTEX A 320,70
70 VERTEX B 540,390
80 VERTEX C 100,390
90 V00D00 TRICOLOR 4096,0,0,4096
100 TRIANGLE
110 V00D00 SWAP
```

Line 40 enables RGB writes and selects the back draw target. Line 90 uses 12.12 fixed-point colour: 4096 means full intensity and 0 means off. The result is a red triangle on a dark background.

9.3 BASIC keywords

The V00D00 keyword introduces most Voodoo subcommands. VERTEX, TRIANGLE, ZBUFFER, and TEXTURE are companion hardware verbs that operate on the same state.

Form	Effect
V00D00 ON / V00D00 OFF	Enable or disable the chip.
V00D00 DIM w,h	Set output dimensions and resize the buffers if valid.
V00D00 CLEAR colour	Store a 32-bit clear word in COLOR0 and trigger FAST_FILL_CMD.
V00D00 SWAP	Flush queued triangles and publish the frame.
V00D00 CLIP x0,y0,x1,y1	Set the scissor rectangle.
V00D00 COMBINE mode	Write FBZCOLOR_PATH. Common values are 0 iterated, 1 texture, 97 modulate.
V00D00 LFB mode	Write LFB_MODE.
V00D00 PIXEL x,y,word	Write one 32-bit word into the linear texture/LFB aperture at $\$D0000 + ((y * width + x) * 4)$.
V00D00 RGB ON / V00D00 RGB OFF	Set or clear the RGB-write bit in FBZ_MODE.
V00D00 TRICOLOR r,g,b[,a]	Write START_R, START_G, START_B, and optional START_A. Values are 12.12 fixed point.
V00D00 TRISHADE drdx,drdy,dgdx,dgdy,dbdx,dbdy	Write colour gradient registers.
V00D00 TRIDEPTH start_z,dzdx,dzdy	Write depth start and gradient registers. Values are 20.12 fixed point.
V00D00 TRIUV start_s,start_t,dsdx,dt dx,dsdy,dt dy	Write texture coordinate start and gradient registers. Values are 14.18 fixed point.

Form	Effect
V00D00 TRIW start_w,dwdx,dwdy	Write perspective W start and gradient registers. Values are 2.30 fixed point.
V00D00 ALPHA TEST ON / OFF	Set or clear alpha-test enable.
V00D00 ALPHA FUNC n	Set alpha-test function bits 1 to 3.
V00D00 ALPHA BLEND ON / OFF	Set or clear alpha-blend enable.
V00D00 ALPHA SRC n / V00D00 ALPHA DST n	Set source and destination blend factors.
V00D00 FOG ON / OFF	Set or clear fog enable.
V00D00 FOG COLOR r,g,b	Set fog colour as an RGB triplet.
V00D00 DITHER ON / OFF	Set or clear ordered dither.
V00D00 CHROMAKEY ON / OFF	Set or clear chroma-key discard.
V00D00 CHROMAKEY COLOR r,g,b	Set the chroma-key colour.
VERTEX A x,y	Write vertex A, converting integer coordinates to 12.4 format.
VERTEX B x,y	Write vertex B.
VERTEX C x,y	Write vertex C.
TRIANGLE	Queue the current triangle.
ZBUFFER ON / OFF	Set or clear depth-test enable.
ZBUFFER FUNC n	Set depth function bits 5 to 7.
ZBUFFER WRITE ON / OFF	Set or clear depth-buffer writes.
TEXTURE ON / OFF	Set or clear texture enable in TEXTURE_MODE.
TEXTURE MODE mode	Write TEXTURE_MODE.
TEXTURE BASE lod,addr	Write one texture base register.
TEXTURE DIM w,h	Set upload width and height.
TEXTURE UPLOAD	Commit texture memory into the texture sampler.

9.4 Data formats

Voodoo registers are 32-bit words. POKE32 writes a whole word. PEEK32 reads a whole word. POKE8 may be used for byte work, but register side effects happen only after byte 3 of a word has been written.

Value	Format	Unit
Vertex X/Y	12.4	1 pixel is 16
R/G/B/A	12.12	full intensity is 4096
Z	20.12	1.0 is 4096
S/T texture coordinate	14.18	1.0 is 262144
W	2.30	1.0 is 1073741824

VERTEX A, VERTEX B, and VERTEX C shift integer coordinates left by four for you. V00D00 TRICOLOR, V00D00 TRIDEPTH, V00D00 TRIUV, and V00D00 TRIW write the raw fixed-point words you supply.

V00D00 CLEAR and COLOR0 use a 32-bit ARGB word: \$AARRGGBB. The texture memory aperture is byte-addressed RGBA. When a 32-bit POKE32 writes one RGBA texel into little-endian texture memory, the word appears as \$AABBGRR. For example, &HFF0000FF stores an opaque red texture pixel.

9.5 Register block

Voodoo's register block runs from \$F8000 to \$F87FF. Registers are 32-bit words at 4-byte aligned addresses.

9.5.1 Control and status

Address	Name	Purpose
\$F8000	V00D00_STATUS	Status.
\$F8004	V00D00_ENABLE	Bit 0 enables the chip.

V00D00_STATUS bits:

Bits	Field	Meaning
0	FBI_BUSY	Framebuffer interface busy.
1	TMU_BUSY	Texture unit busy.
2	SST_BUSY	Overall chip busy.
6	VRETRACE	Vertical retrace active.
7	SWAPBUF	Buffer swap pending.
12-19	MEMFIFO	Coarse memory FIFO free-space field.
20-24	PCIFIFO	Command FIFO free-space field.

MEMFIFO is useful as a ready/not-ready field, not as a cycle-accurate historical FIFO depth. It is non-zero while the queued triangle batch can accept more triangles. It reads zero once the batch reaches its 4096 triangle limit. PCIFIFO reports command space for the current high-level engine path.

FBI_BUSY and SST_BUSY are set while SWAP_BUFFER_CMD is flushing queued triangles, swapping buffers, and publishing the frame. A simple BASIC program may not see these bits because the command completes before the next PEEK, but machine code and IE Mon can still poll them around a large swap. TRIANGLE_CMD itself does not set these busy bits; it only appends work to the batch.

9.5.2 Vertex and attribute registers

Address	Name	Format
\$F8008 to \$F801C	VERTEX_AX to VERTEX_CY	12.4 fixed point
\$F8020	START_R	12.12 fixed point
\$F8024	START_G	12.12 fixed point
\$F8028	START_B	12.12 fixed point
\$F802C	START_Z	20.12 fixed point

Address	Name	Format
\$F8030	START_A	12.12 fixed point
\$F8034	START_S	14.18 fixed point
\$F8038	START_T	14.18 fixed point
\$F803C	START_W	2.30 fixed point
\$F8040 to \$F805C	DRDX to DWDX	per-X gradients
\$F8060 to \$F807C	DRDY to DWDY	per-Y gradients

Without `COLOR_SELECT`, every submitted triangle uses the current start values for all three vertices. Writing `COLOR_SELECT` with 0, 1, or 2 selects vertex A, B, or C as the target for later `START_*` writes. The selection applies to colour, Z, S, T, and W. After `TRIANGLE`, Gouraud selection is reset for the next triangle.

9.5.3 Command registers

Address	Name	Purpose
\$F8080	TRIANGLE_CMD	Queue one triangle using current state.
\$F8084	FTRIANGLECMD	Queue one triangle through the fast-triangle alias.
\$F8088	COLOR_SELECT	Select vertex 0, 1, or 2 for later attribute writes.
\$F8120	NOP_CMD	No operation.
\$F8124	FAST_FILL_CMD	Clear the drawing buffer with <code>COLOR0</code> .
\$F8128	SWAP_BUFFER_CMD	Flush triangles and publish the frame. Bit 0 waits for retrace; bit 1 clears after swap.

`TRIANGLE_CMD` queues work. It does not rasterise visible pixels and it does not wait for the rasteriser to draw the triangle. The current vertex and attribute state is copied into the triangle batch, up to 4096 triangles. If the batch is already full, further `TRIANGLE_CMD` writes are ignored until a swap flushes the batch.

Pixels appear after `SWAP_BUFFER_CMD`. That command updates dirty pipeline state, flushes the queued triangles into the Voodoo backend, clears the batch, swaps buffers, and publishes the frame to the compositor. During that flush the status register reports `FBI_BUSY` and `SST_BUSY`.

9.5.4 Mode and state

Address	Name	Purpose
\$F8104	FBZCOLOR_PATH	Colour combine and source selects.
\$F8108	FOG_MODE	Fog pipeline.
\$F810C	ALPHA_MODE	Alpha test and alpha blend.
\$F8110	FBZ_MODE	Framebuffer, Z, clipping, and draw enable.
\$F8114	LFB_MODE	Linear framebuffer mode latch.
\$F8118	CLIP_LEFT_RIGHT	(left << 16) right.
\$F811C	CLIP_LOW_Y_HIGH	(top << 16) bottom.

`FBZ_MODE` bits:

Bit	Field	Meaning
0	CLIPPING	Enable scissor.
1	CHROMAKEY	Enable chroma-key discard.
2	STIPPLE	Enable stipple mask.
3	WBUFFER	Use W-buffer interpretation for depth normalisation.
4	DEPTH_ENABLE	Enable depth test.
5-7	DEPTH_FUNC	Depth compare function.
8	DITHER	Enable ordered dither.
9	RGB_WRITE	Enable RGB writes.
10	DEPTH_WRITE	Enable depth writes.
11	DITHER_2X2	Use 2x2 dither instead of 4x4.
12	ALPHA_WRITE	Enable alpha buffer writes.
14	DRAW_FRONT	Draw to front buffer.
15	DRAW_BACK	Draw to back buffer.
16	DEPTH_SOURCE	Select depth source.
17	Y_ORIGIN	Use bottom-left Y origin when set.
18	ALPHA_PLANES	Preserve interpolated alpha in the target.
19	ALPHA_DITHER	Dither alpha.
20-31	DEPTH_OFFSET	Depth bias.

The default FBZ_MODE enables RGB, depth test, depth write, and depth function LESS. If neither draw-target bit is set, the normal back draw buffer is used.

Depth and alpha function numbers:

Value	Function
0	NEVER
1	LESS
2	EQUAL
3	LESSEQUAL
4	GREATER
5	NOTEQUAL
6	GREATEREQUAL
7	ALWAYS

ALPHA_MODE bits:

Bit	Field	Meaning
0	TEST_EN	Enable alpha test.

Bit	Field	Meaning
1-3	TEST_FUNC	Alpha test function.
4	BLEND_EN	Enable alpha blending.
5	ANTIALIAS	Antialias flag.
8-11	SRC_RGB	Source RGB blend factor.
12-15	DST_RGB	Destination RGB blend factor.
16-19	SRC_A	Source alpha blend factor.
20-23	DST_A	Destination alpha blend factor.
24-31	REF	Alpha-test reference value.

Blend-factor numbers:

Value	Factor
0	zero
1	source alpha
2	colour average
3	destination alpha
4	one
5	one minus source alpha
6	one minus colour average
7	one minus destination alpha
15	source-alpha saturate

9.6 Fog, chroma, stipple, and constants

Address	Name	Purpose
\$F8140 to \$F823C	FOG_TABLE_BASE	64 fog-table entries.
\$F81C4	FOG_COLOR	Fog RGB.
\$F81C8	ZA_COLOR	Z/A constant.
\$F81CC	CHROMA_KEY	Chroma-key colour.
\$F81D0	CHROMA_RANGE	Chroma-key range.
\$F81D4	STIPPLE	Stipple pattern.
\$F81D8	COLOR0	Constant colour 0, used by fast fill.
\$F81DC	COLOR1	Constant colour 1.

`V00D00 FOG COLOR r,g,b` and `V00D00 CHROMAKEY COLOR r,g,b` pack the triplet as $(b \ll 16) \mid (g \ll 8) \mid r$. Fog blends the triangle colour towards the fog colour by the fragment Z value clamped to 0.0 to 1.0. Chroma key discards a pixel if the final RGB colour matches the key within a one-count tolerance; if `CHROMA_RANGE` is non-zero, the key becomes the low RGB bound and range becomes the high RGB bound.

Stipple uses the low 32 bits as a 4 row by 8 column mask. A zero stipple word lets every pixel through.

9.7 Texture unit

Address	Name	Purpose
\$F8300	TEXTURE_MODE	Texture mode and format.
\$F8304	TLOD	Level-of-detail control.
\$F8308	TDETAIL	Detail texture control.
\$F830C to \$F832C	TEX_BASE0 to TEX_BASE8	Base address per level.
\$F8330	TEX_WIDTH	Upload width.
\$F8334	TEX_HEIGHT	Upload height.
\$F8338	TEX_UPLOAD	Commit texture memory to the sampler.
\$F8400 to \$F87FF	PALETTE_BASE	256 texture palette entries.

TEXTURE_MODE bits:

Bit	Field	Meaning
0	ENABLE	Enable texture sampling.
1-3	MINIFY	Minification filter.
4	MAGNIFY	Magnification filter.
5	CLAMP_S	Clamp S to 0.0 through 1.0.
6	CLAMP_T	Clamp T to 0.0 through 1.0.
8-11	FORMAT	Texture format.
12	CHROMA	Texture chroma key flag.
13	TRILINEAR	Trilinear flag.
14	PERSPECTIVE	Perspective-correct S/T interpolation.
15	DETAIL	Detail texture flag.
16	SEQUENCE	Sequence flag.

Texture format numbers:

Code	Format
0	8-bit palette
1	YIQ
2	A8
3	I8
4	AI44
5	P8
6	ARGB8332

Code	Format
7	AI88
8	ARGB1555
9	ARGB4444
10	ARGB8888

Upload sequence:

1. Write texels into \$D0000.
2. Set TEXTURE DIM w,h.
3. Set TEXTURE MODE.
4. Use TEXTURE UPLOAD.
5. Enable texturing with TEXTURE ON.
6. Select a texture combine path with V00D00 COMBINE.

Upload is ignored if width or height is zero, or if $w * h * 4$ exceeds 64 KB.

9.8 Colour pipeline

FBZCOLOR_PATH selects the local colour, the other colour, and the combine function used when texturing is active.

Bits	Field	Meaning
0-1	RGB source	0 iterated, 1 texture, 2 COLOR1, 3 LFB.
2-3	Alpha source	Same encoding as RGB.
4-6	Combine mode	One of the combine functions below.
27	Texture enable	Include texture colour in the path.

Combine code	Function	Meaning
0	ZERO	Output black.
1	CSUB_CL	Other minus local.
2	ALOCAL	Local times local alpha.
3	AOTHER	Local times other alpha.
4	CLOCAL	Local pass through.
5	ALOCAL_T	Local alpha times texture.
6	CLOC_MUL	Local times other.
7	AOTHER_T	Other alpha times texture.

Convenience values accepted by V00D00 COMBINE:

Value	Meaning
0	Iterated vertex colour.
1	Texture colour.
97	Texture times vertex colour.

9.9 Examples

9.9.1 Gouraud colour fan

COLOR_SELECT lets a program write a different colour at each vertex. This produces a smooth red, green, and blue triangle.

```
10 REM V00D00 GOURAUD FAN
20 V00D00 ON
30 V00D00 DIM 640,480
40 POKE32 &H000F8110,&H00008200
50 V00D00 CLEAR &HFF000018
60 VERTEX A 320,60
70 VERTEX B 560,410
80 VERTEX C 80,410
90 POKE32 &H000F8088,0
100 V00D00 TRICOLOR 4096,0,0,4096
110 POKE32 &H000F8088,1
120 V00D00 TRICOLOR 0,4096,0,4096
130 POKE32 &H000F8088,2
140 V00D00 TRICOLOR 0,0,4096,4096
150 TRIANGLE
160 V00D00 SWAP
```

Expected result: a large triangle blends smoothly between red at the top, green at the lower right, and blue at the lower left.

9.9.2 Depth-tested overlap

This draws a far blue triangle first and a nearer yellow triangle second. With depth function LESS, the nearer triangle wins where the two overlap.

```
10 REM V00D00 Z OVERLAP
20 V00D00 ON
30 V00D00 DIM 640,480
40 POKE32 &H000F8110,&H00008630
50 V00D00 CLEAR &HFF000000
60 VERTEX A 220,80
70 VERTEX B 520,390
80 VERTEX C 80,390
90 V00D00 TRICOLOR 0,0,4096,4096
100 V00D00 TRIDEPTH 3277,0,0
110 TRIANGLE
120 VERTEX A 330,120
130 VERTEX B 560,350
140 VERTEX C 160,350
150 V00D00 TRICOLOR 4096,4096,0,4096
160 V00D00 TRIDEPTH 819,0,0
170 TRIANGLE
180 V00D00 SWAP
```

Line 40 sets DRAW_BACK, DEPTH_WRITE, RGB_WRITE, DEPTH_ENABLE, and depth function LESS. 3277 is about 0.8 in 20.12 fixed point. 819 is about 0.2.

9.9.3 Texture upload and textured triangle

This builds a four-pixel texture in texture memory and maps it across a triangle.

```

10 REM V00D00 2 BY 2 TEXTURE
20 V00D00 ON
30 V00D00 DIM 640,480
40 POKE32 &H000F8110,&H00008200
50 V00D00 CLEAR &HFF080808
60 POKE32 &H000D0000,&HFF0000FF
70 POKE32 &H000D0004,&HFF00FF00
80 POKE32 &H000D0008,&HFFFF0000
90 POKE32 &H000D000C,&HFFFFFFF
100 TEXTURE DIM 2,2
110 TEXTURE MODE &H0A61
120 TEXTURE UPLOAD
130 TEXTURE ON
140 V00D00 COMBINE 1
150 VERTEX A 320,60
160 VERTEX B 570,410
170 VERTEX C 70,410
180 POKE32 &H000F8088,0
190 V00D00 TRICOLOR 4096,4096,4096,4096
200 V00D00 TRIUV 0,0,0,0,0,0
210 POKE32 &H000F8088,1
220 V00D00 TRICOLOR 4096,4096,4096,4096
230 V00D00 TRIUV 262144,0,0,0,0,0
240 POKE32 &H000F8088,2
250 V00D00 TRICOLOR 4096,4096,4096,4096
260 V00D00 TRIUV 0,262144,0,0,0,0
270 TRIANGLE
280 V00D00 SWAP

```

&H0A61 enables texturing, selects ARGB8888 format, and clamps S and T. The texture words are written as little-endian RGBA texels. Expected result: the triangle samples red, green, blue, and white texels.

9.9.4 Linear aperture inspection

V00D00 PIXEL writes a 32-bit word into the texture/LFB aperture. It is useful for building texture memory and for checking address maths.

```

10 REM V00D00 PIXEL INSPECT
20 V00D00 ON
30 V00D00 DIM 320,200
40 V00D00 PIXEL 10,5,42
50 PRINT PEEK32(&H000D1928)

```

The printed value is 42, because $(5 * 320 + 10) * 4$ is 6440, and $\$D0000 + 6440$ is $\$D1928$.

9.9.5 Alpha, fog, chroma key, and dither

This draws translucent fogged colour over a green keyed triangle. The green triangle is discarded by the chroma key; the magenta triangle is drawn with alpha blending and dither.

```

10 REM V00D00 PIPELINE FLAGS
20 V00D00 ON
30 V00D00 DIM 640,480
40 POKE32 &H000F8110,&H00048202
50 V00D00 CLEAR &HFF202020
60 V00D00 CHROMAKEY COLOR 0,255,0
70 V00D00 CHROMAKEY ON
80 V00D00 DITHER ON
90 V00D00 FOG COLOR 40,40,80
100 V00D00 FOG ON
110 POKE32 &H000F810C,&H00005110
120 VERTEX A 320,80
130 VERTEX B 540,390
140 VERTEX C 100,390
150 V00D00 TRICOLOR 0,4096,0,4096
160 V00D00 TRIDEPTH 3000,0,0
170 TRIANGLE
180 V00D00 CHROMAKEY OFF
190 VERTEX A 330,110
200 VERTEX B 520,360
210 VERTEX C 140,360
220 V00D00 TRICOLOR 4096,0,4096,2048
230 V00D00 TRIDEPTH 1800,0,0
240 TRIANGLE
250 V00D00 SWAP

```

Line 40 keeps alpha planes so the 2048 alpha on line 220 reaches the blend unit. Line 110 sets blend enable, source factor SRC_ALPHA, and destination factor ONE_MINUS_SRC_ALPHA. Expected result: the green triangle leaves the cleared background untouched, while the magenta triangle appears softened by fog, alpha, and dither.

9.10 Side effects and limits

Voodoo has these programming boundaries:

Action	Side effect or limit
V00D00 OFF	Voodoo contributes no picture to the compositor.
V00D00 DIM w, h	Ignored unless both dimensions are positive and no larger than 800 by 600.
V00D00 CLEAR colour	Clears the drawing buffer and resets depth to a far value for LESS style depth functions.
TRIANGLE	Queues one triangle, up to 4096 queued triangles; extra submissions are ignored until swap.
V00D00 SWAP	Flushes queued triangles, sets busy while the flush runs, swaps buffers, publishes the frame, and clears the batch.
SWAP_BUFFER_CMD bit 1	Clears the drawing buffer after the swap using current COLOR0.
POKE8 to registers	Updates the shadow byte immediately, but command side effects run only when byte 3 of the word is written.
Texture upload	Copies $w * h * 4$ bytes from \$D0000 if the size fits in 64 KB.
Texture sampling	Uses point sampling, with wrap by default and clamp when CLAMP_S or CLAMP_T is set.
Chroma key	Discards a final pixel colour that matches the key or keyed range.
Fog	Blends by the clamped Z value.

Action	Side effect or limit
Dither	Quantises RGB through an ordered matrix before writing.
Draw targets	Front, back, or both may be selected; if no target bit is set, the normal back draw buffer is used.

Degenerate triangles with zero area are ignored. The rasteriser clips to the picture bounds and then to the scissor rectangle when clipping is enabled. Back-facing triangles are reordered internally so that clockwise and anticlockwise vertex order both draw.